

Music Observatory: project goal & complete flow

An explorable mathematical account of how acoustic signals relate to musical structure and human perception.

Project goal

Build an interactive mathematical and statistical observatory for music. Represent a recording at several scales, visualize low-level acoustic measurements alongside mid-level musical organization, and investigate their relationships with high-level perceptual descriptors.

The central question is: How much of the structure people perceive in music can be recovered, explained, and predicted from the statistical structure of its acoustic signal?

The project serves your interest in music, mathematics, and creative explanation. Its research value comes from feature engineering, inference, experimental design, and interpretable analysis. Financial prediction is outside its scope.

Complete analytical flow

Stage	Input and operation	Output	User question
1. Ingest	Decode audio; record sample rate, channels, duration and provenance	Waveform and metadata	What exactly am I analyzing?
2. Prepare	Channel policy, optional resampling, frames, windows and silence masks	Aligned analysis frames	What do preprocessing choices change?
3. Transform	STFT, mel filterbanks; later CQT or wavelets	Time-frequency representations	Where do frequencies and transients occur?
4. Measure	Amplitude, spectrum, timbre, onset and pitch features	Descriptor matrix $D \in \mathbb{R}^{T \times d}$	What changes acoustically?
5. Organize	Beat synchronization, scaling, covariance, PCA and similarity	Trajectories and recurrent patterns	Which properties move together?
6. Infer structure	Novelty, change points, clustering; later HMMs	Boundary candidates and latent states	Which moments repeat or mark transitions?
7. Relate to perception	Human annotations, regression and probabilistic models	Predictions with evaluated uncertainty	What measurements explain listener judgments?
8. Experiment	Controlled transformations, held-out evaluation and population statistics	Effect sizes, intervals and limitations	Is the apparent relationship robust?

These stages form a dependency graph, not a proof of causation. Rhythm and harmony branch from the transforms and rejoin multivariate analysis; perceptual labels come from listeners or a documented external model, rather than automatically from unsupervised clusters.

Descriptor hierarchy

Low level: RMS, peak amplitude, crest factor, zero-crossing rate, centroid, spread, rolloff, flatness, entropy, flux, MFCCs and onset strength. Chroma is a pitch-oriented representation often placed near the low/mid-level boundary.

Mid level: tempo hypotheses, beats, rhythmic patterns, pitch contours, chord candidates, harmonic movement, repetition and section boundaries. Instrumentation and melody extraction require additional estimation methods; they are extensions, not consequences of spectral moments alone.

High level: listener-rated energy, brightness, tension, danceability and mood; semantic labels such as verse, chorus and drop. These concepts depend on listener, culture, context and annotation protocol.

Rhythm complexity and harmonic stability need explicit operational definitions before they become measurable targets.

Exploration flow

1. Load a recording and choose an analysis resolution.
2. Play it with a shared time cursor across waveform, spectrogram, feature curves, chroma, novelty and detected events.
3. Select an interval to inspect its distributions, covariance and nearest recurring moments.
4. Follow a descriptor down to its formula, transform, window and samples; follow it upward to associated structure and annotated perception.
5. Compare alternative settings or transformed audio, then save an observation with its parameters.
6. Run a declared experiment across recordings and inspect uncertainty and held-out performance.

A high-energy label might be associated with loudness, onset density and spectral brightness. The interface should expose those associations and their uncertainty; it must not present them as a universal or causal explanation.

Knowledge map

- Signal processing and low-level statistics
- Pitch, harmony, rhythm and psychoacoustics
- Multivariate structure, information and similarity
- Temporal models and musical segmentation
- Inference, perception and cross-song experiments
- Implementation roadmap and validation

Intended outcome

A working analysis laboratory with synchronized audio, transparent feature definitions, reproducible analysis settings, and an experimental workflow. Success means a user can inspect a musical event, understand its measurements, and distinguish a measured quantity from an inferred or listener-dependent interpretation.